

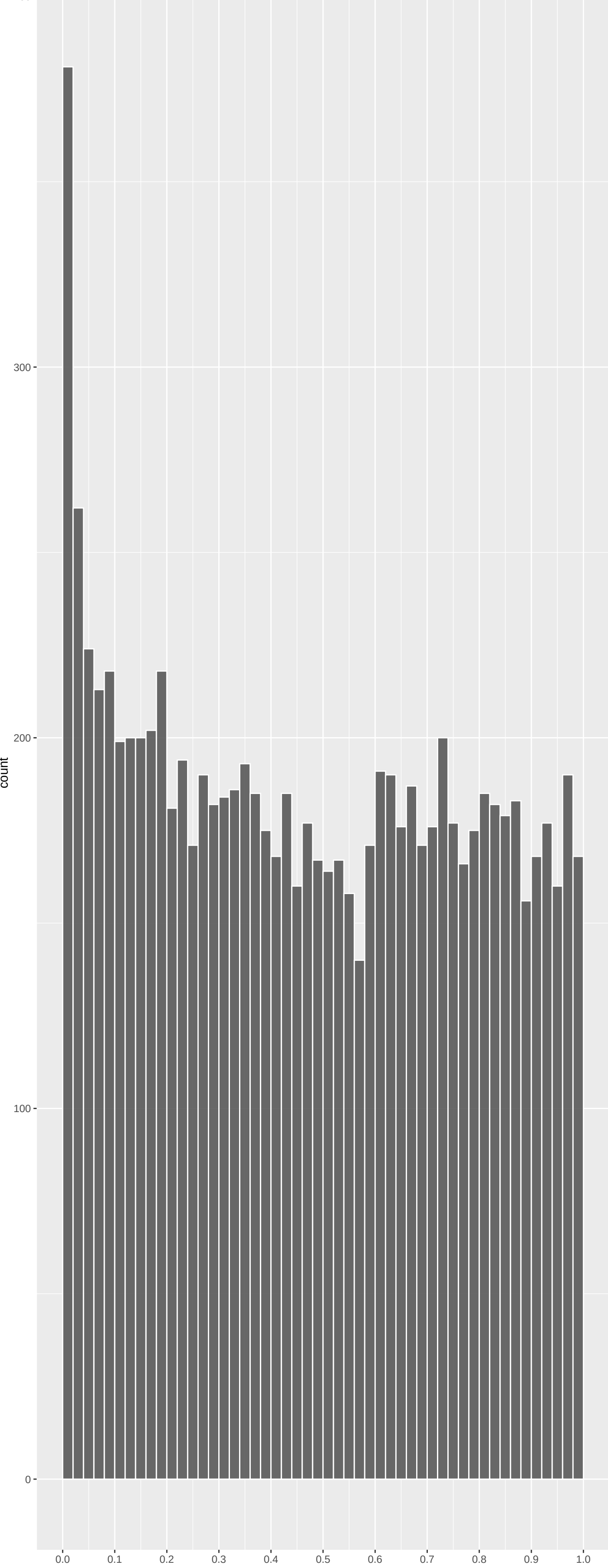
Top genes by P-value non-permuted

Gene	Rho	N	P	p.adj	qValueNoperm
GTSF1	0.5195278	90	1.546e-07	2.921e-03	1.000e+00
OR51E1	0.3769524	127	1.252e-05	1.183e-01	1.000e+00
MAP3K4	-0.3078446	166	5.466e-05	2.878e-01	1.000e+00
MYH1	0.3244522	147	6.092e-05	2.878e-01	1.000e+00
ACTN4	0.3281964	103	7.133e-04	4.093e-01	1.000e+00
ASIC4	0.3041215	126	5.359e-04	4.093e-01	1.000e+00
ATP5MF-PTCD1	0.2710065	163	4.669e-04	4.093e-01	1.000e+00
CELSR2	0.2912994	158	2.045e-04	4.093e-01	1.000e+00
COL8A2	0.2857188	140	6.217e-04	4.093e-01	1.000e+00
DYNLL2	-0.6577996	24	4.768e-04	4.093e-01	1.000e+00
HI-3	0.3042161	137	3.013e-04	4.093e-01	1.000e+00
HOXB8	0.4799820	46	7.366e-04	4.093e-01	1.000e+00
IRAK1	-0.2987448	142	3.046e-04	4.093e-01	1.000e+00
KIF5C	0.3339514	103	5.660e-04	4.093e-01	1.000e+00
KLHL3	0.3316369	104	5.839e-04	4.093e-01	1.000e+00
MANSC4	-0.2722302	153	6.635e-04	4.093e-01	1.000e+00
MEPE	-0.2759656	147	7.162e-04	4.093e-01	1.000e+00
MPP2	0.3365229	105	4.481e-04	4.093e-01	1.000e+00
MYH4	0.2746720	151	6.424e-04	4.093e-01	1.000e+00
NLGN3	0.3568919	87	6.908e-04	4.093e-01	1.000e+00
OR51S1	0.2890290	145	4.222e-04	4.093e-01	1.000e+00
OR5AR1	0.3089518	116	7.124e-04	4.093e-01	1.000e+00
PDCD5	0.4404200	63	3.036e-04	4.093e-01	1.000e+00
POU3F4	0.3497440	92	6.323e-04	4.093e-01	1.000e+00
POU4F3	0.4510655	56	4.849e-04	4.093e-01	1.000e+00

Top genes by Q-Value non-permuted

Gene	Rho	N	P	p.adj	qValueNoperm
A1BG	0.067210856	125	4.564e-01	9.376e-01	1.000e+00
A1CF	-0.002108205	131	9.809e-01	9.988e-01	1.000e+00
A2M	-0.103833605	173	1.740e-01	8.506e-01	1.000e+00
A2ML1	0.072528856	136	4.014e-01	9.231e-01	1.000e+00
A3GALT2	0.063881486	136	4.600e-01	9.379e-01	1.000e+00
A4GALT	-0.205419100	156	1.010e-02	5.794e-01	1.000e+00
AAGNT	-0.052469229	155	5.167e-01	9.475e-01	1.000e+00
AAAS	0.040545818	136	6.989e-01	9.688e-01	1.000e+00
AACS	0.112016737	159	1.598e-01	8.429e-01	1.000e+00
ADAC	-0.098915805	111	3.017e-01	9.018e-01	1.000e+00
AADACL2	0.118539595	123	1.916e-01	8.601e-01	1.000e+00
AADACL3	0.020817112	162	7.926e-01	9.826e-01	1.000e+00
AADACL4	0.019361279	145	8.172e-01	9.831e-01	1.000e+00
ADAT	0.018927524	154	8.158e-01	9.831e-01	1.000e+00
AGAB	-0.097616452	139	2.529e-01	8.863e-01	1.000e+00
AAK1	0.201697998	148	1.396e-02	5.856e-01	1.000e+00
AAMDC	0.115676265	89	2.804e-01	8.944e-01	1.000e+00
AAMP	0.299142641	98	2.770e-03	5.113e-01	1.000e+00
AANAT	-0.086466994	142	3.062e-01	9.018e-01	1.000e+00
AAR2	0.106323873	137	2.162e-01	8.723e-01	1.000e+00
AARD	-0.059017699	109	5.421e-01	9.533e-01	1.000e+00
AARS1	0.018662501	148	8.219e-01	9.834e-01	1.000e+00
AARS2	0.062062751	169	4.228e-01	9.274e-01	1.000e+00
AARSD1	0.075080631	148	3.644e-01	9.131e-01	1.000e+00
AASDH	-0.059911042	169	4.391e-01	9.321e-01	1.000e+00

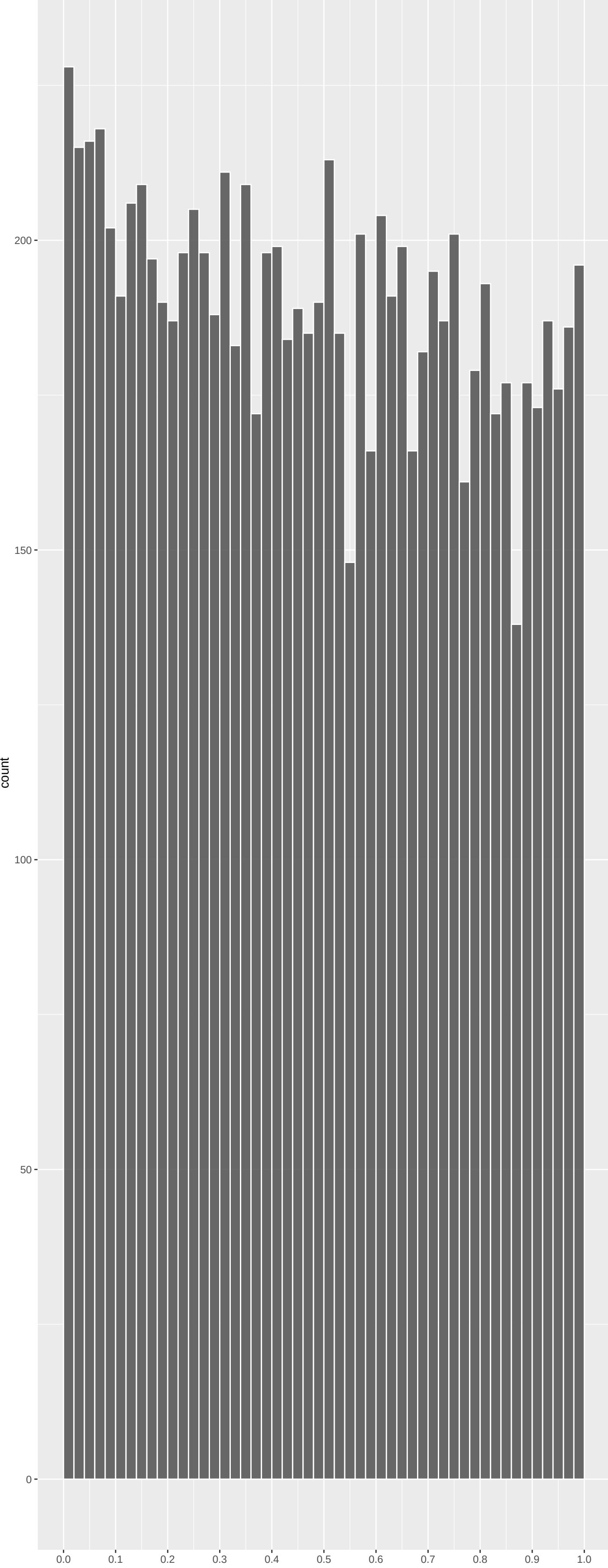
Positive Rho Non-permuted



Top Positive genes by P-value non-permuted

Gene	Rho	N	P	p.adj	qValueNoperm
GTSF1	0.5195278	90	1.546e-07	2.921e-03	1.000e+00
OR51E1	0.3769524	127	1.252e-05	1.183e-01	1.000e+00
MYH1	0.3244522	147	6.092e-05	2.878e-01	1.000e+00
ACTN4	0.3281964	103	7.133e-04	4.093e-01	1.000e+00
ASIC4	0.3041215	126	5.359e-04	4.093e-01	1.000e+00
ATP5MF-PTCD1	0.2710065	163	4.669e-04	4.093e-01	1.000e+00
CELSR2	0.2912994	158	2.045e-04	4.093e-01	1.000e+00
COL8A2	0.2857188	140	6.217e-04	4.093e-01	1.000e+00
HI-3	0.3042161	137	3.013e-04	4.093e-01	1.000e+00
HOXB8	0.4799820	46	7.366e-04	4.093e-01	1.000e+00

Negative Rho Non-permuted



Top Negative genes by P-value non-permuted

Gene	Rho	N	P	p.adj	qValueNoperm
MAP3K4	-0.3078446	166	5.466e-05	2.878e-01	1.000e+00
DYNLL2	-0.6577996	24	4.768e-04	4.093e-01	1.000e+00
IRAK1	-0.2987448	142	3.046e-04	4.093e-01	1.000e+00
MANSC4	-0.2722302	153	6.635e-04	4.093e-01	1.000e+00
MEPE	-0.2759656	147	7.162e-04	4.093e-01	1.000e+00
ZNF875	-0.3455472	96	5.644e-04	4.093e-01	1.000e+00
BMP2K	-0.2588516	161	9.146e-04	4.320e-01	1.000e+00
TAS2R9	-0.3190597	105	9.086e-04	4.320e-01	1.000e+00
ADAMTS7	-0.2557626	158	1.181e-03	4.334e-01	1.000e+00
C18ORF63	-0.2538905	164	1.036e-03	4.334e-01	1.000e+00

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Cytoplasmic Translation (GO:0002181)	0.19366448	91	1.784e-10	9.648e-07	RPS8:11 RPS3:46 RPS17:48 RPL27:167 RPL32:231 RPS23:343
Chemical Synaptic Transmission (GO:00072	0.10482732	266	4.458e-09	1.205e-05	NLGN3:24 RAPSN:38 GRIA2:70 GABRB3:92 DOC2A:107 CACNB3:118
Macromolecule Biosynthetic Process (GO:0	0.12904556	179	1.291e-08	2.172e-05	RPS8:11 RPS3:46 RPS17:48 RPL27:167 MRPL3:188
Monatomic Cation Transmembrane Transpor	0.09897257	277	1.607e-08	2.172e-05	ASIC4:17 ATP1A2:40 KCNS1:116 KCNS2:155 KCNH6:199 RYR3:212
Peptide Biosynthetic Process (GO:0043043)	0.13162592	153	2.033e-08	2.199e-05	RPS8:11 RPS3:46 RPS17:48 RPL27:167 MRPL3:188
Brain Development (GO:0007420)	0.12586620	157	5.555e-08	4.290e-05	ZIC1:22 PITPNM1:102 GRIN2C:308 GIT1:350 GRIN2D:519 CDK5R1:550
Gene Expression (GO:0010467)	0.09370695	286	5.523e-08	4.290e-05	RPS8:11 RPS3:46 RPS17:48 KPNAA:495 ALYREF:311 MAGOH:146
Interstrand Cross-Link Repair (GO:003629	-0.25987203	36	6.881e-08	4.651e-05	FANCB:35 FANCF:160 FANCD2:321 RNF168:424 FANCI:716 DCLRE1A:781
Inflammatory Response (GO:0006954)	-0.10277885	229	9.104e-08	4.923e-05	ADM:38 PTGER4:96 LIAS:98 ODM3:311 LOXL3:491 RELA:506
Signal Release From Synapse (GO:0099643)	0.25410818	37	8.933e-08	4.923e-05	STX1A:127 SNPH:134 PPFIA3:138 PDZD11:143 STXBP1:312 STXBP2:557
Canonical Wnt Signaling Pathway (GO:0060	0.19356361	63	1.096e-07	5.387e-05	WNT3:4 WNT1:42 WNT10A:147 SFRP1:406 FZD9:567 FZD3:638
Modulation Of Chemical Synaptic Transmis	0.13990466	118	1.585e-07	7.144e-05	NLGN3:24 ATP1A2:40 GRIA2:70 GRK13:29 APP:209 SLC6A9:313
Cytokine-Mediated Signaling Pathway (GO:0	-0.09568222	249	2.175e-07	9.046e-05	IRAK1:2 TRADD:20 ADIPOR2:195 TNFRSF11A:204 IL27RA:232 IL6R:262
Actomyosin Structure Organization (GO:GO:	0.16893290	77	3.041e-07	1.142e-04	MYH7B:37 SRC:89 TNNT3:178 PDCD10:317 CAPN3:439 MYL9:443
Translation (GO:0006412)	0.09855428	228	3.169e-07	1.142e-04	RPS8:11 RPS3:46 RPS17:48 RPL27:167 MRPL3:15174 RPL3L:188
Neurotransmitter Secretion (GO:0007269)	0.122687802	42	3.664e-07	1.238e-04	DOC2A:107 STX1A:127 SNPH:134 PPFIA3:138 PDZD11:143 STXBP1:312
RNA Splicing, Via Transcription Factor Re	0.11204440	171	4.547e-07	1.384e-04	SNRPD:198 MAGOH:146 SNRPF:175 LSM5:189 SRSF1:227 DXH38:230
Vesicle-Mediated Transport (GO:0016192)	0.07374169	401	4.607e-07	1.384e-04	SPTBN2:36 CSNK1G3:44 DNMI:64 SNX18:93 STX18:96 DOC2A:107
mRNA Processing (GO:0006397)	0.10109558	205	6.442e-07	1.833e-04	CPSF6:87 SNRPD:198 ALYREF:311 MAGOH:146 SNRPF:175 LSM5:189
Inorganic Cation Transmembrane Transport	0.08520202	280	1.011e-06	2.735e-04	ASIC4:17 ATP1A2:40 KCNS1:116 KCNS2:155 KCNH6:199 RYR3:212
Synaptic Vesicle Cycle (GO:0099504)	0.22700630	37	1.780e-06	4.584e-04	STX1A:127 GIT1:350 CDK5:764 SNAP25:1015 OTOF:1114 RAB3A:1138
Metal Ion Transport (GO:0030001)	0.10579373	169	2.185e-06	5.138e-04	RYR1:10 ATP1A2:40 KCNS1:116 CACNB3:118 KCNS2:155 RYR3:212
Positive Regulation Of Cytokine Producti	-0.07892042	307	2.162e-06	5.138e-04	IRAK1:2 ILNRP9:10 BRCA1:33 IL16:87 PTGER4:96 IRF1:126
Regulation Of Neuronal Synaptic Plastici	0.23564769	33	2.818e-06	6.348e-04	PPFIA3:138 NCDN:157 APP:209 GRIN2C:308 EPHB2:410 CAMK2A:467
DNA Repair (GO:0006281)	-0.08094890	293	3.103e-06	6.712e-04	BRCA1:33 FANCB:35 PARRP:240 ZGRF1:104 BARD1:107 XRC2:117
Central Nervous System Development (GO:0	0.08232231	268	3.768e-06	7.823e-04	ZIC1:22 PAX6:56 PITPNM1:102 APP:209 NHLH2:253 GRIN2C:308
mRNA Splicing, Via Spliceosome (GO:00003	0.09467499	201	3.906e-06	7.823e-04	SNRPD:198 MAGOH:146 SNRPF:175 LSM5:189 SRSF1:227 DXH38:230
Protein Transport (GO:0015031)	0.07554185	308	5.572e-06	1.076e-03	SCAMP2:58 SNX18:93 STX1B:96 STX1A:127 RAB38:135 RAB1A:228
Cellular Response To Cytokine Stimulus (	-0.07552792	293	7.112e-06	1.279e-03	ADAMT57:13 TRADD:20 BRCA1:33 IRF1:126 PGM5:51 IL27RA:232
Regulation Of Monoatomic Cation Transpor	0.20755937	39	7.331e-06	1.279e-03	GNB2:75 GRIN2C:308 GRIN2D:519 CACNA1D:541 RGSN:654 KCNP2:674
Replication Fork Processing (GO:0031297)	-0.19344733	45	7.190e-06	1.279e-03	BRCA1:33 BOD1L1:50 GENI:84 BARD1:107 BRCA2:201 FANCM:860
Sensory Perception Of Smell (GO:0007608)	0.08976951	208	8.525e-06	1.440e-03	OR2C1:62 OR2S2:66 OR1021:132 OMP:234 OR6K2:250 OR6N1:271
Wnt Signaling Pathway (GO:0016055)	0.14955693	74	8.796e-06	1.441e-03	WNT3:4 CELSR2:6 WNT1:42 WNT10A:147 SFRP1:406 FZD9:567
Homologous Chromosome Pairing At Meiosis	-0.23100291	30	1.197e-05	1.903e-03	TERB1:59 SYCP2:73 FANCD2:321 IHO1:508 SHOC1:542 RNF218:721
Actin-Myosin Filament Sliding (GO:003327	0.38011315	11	1.269e-05	1.907e-03	MYH4:23 MYH2:22 MYL1:254 MYL6:1040 TNNT2:1575 TPM1:1718
Striated Muscle Contraction (GO:0006941)	0.16892606	56	1.241e-05	1.907e-03	RYR1:10 MYH7B:37 ATP1A2:40 TNNC2:52 TNNT3:178 RYR3:212
Axon Development (GO:0061564)	0.12743083	97	1.474e-05	2.098e-03	MAP1B:35 SSNA1:202 APP:209 OLIG2:372 SPTBN4:374 ANK3:395
Muscle Contraction (GO:0006936)	0.13091600	92	1.454e-05	2.098e-03	MYH1:3 RYR1:10 MYH4:23 MYH13:111 RYR3:212 MYH2:22
Protein Localization To Cell Periphery (	0.10669296	137	1.671e-05	2.317e-03	FLOT1:43 CACNB3:118 RAB38:135 PDZD11:143 RAP2A:226 STXBP1:312
Potassium Ion Transport (GO:0006813)	0.11309172	121	1.781e-05	2.407e-03	ATP1A2:40 KCNS1:116 KCNS2:155 KCNH6:199 LRRRC38:233 KCNJ8:490

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_DEVELOPMENTAL_BIOLOGY	0.11576820	1143	2.598e-39	1.687e-35	RPS8:11 SPTBN2:36 WNT1:42 RPS3:46 RPS17:48 PIK3R1:55
HOUNKPE_HOUSEKEEPING_GENES	0.11477461	1100	2.268e-37	7.366e-34	MSL2:60 AAMP:69 PSMC1A:82 PCBP2:85 KPNAA:495 EIF2S1:100
BLALOCK_ALZHEIMERS_DISEASE_DN	0.10193746	1211	1.793e-32	3.882e-29	PDCD5:9 KIF5C:18 NIPSNAP1:32 HPCA:33 ATP1A2:40 CSNK1G3:44
KIM_BIPOLAR_DISORDER_OLIGODENDROCYTE_DEN	0.13139622	655	2.835e-30	4.603e-27	HPCA:33 SPTBN2:36 HSPA2:39 RPS17:48 SV2A:54 DNMI:64
KIM_ALL_DISORDERS_OLIGODENDROCYTE_NUMBER	0.12268962	729	2.626e-29	3.411e-26	RPS8:11 NIPSNAP1:32 HPCA:33 SPTBN2:36 HPCA:33 RPS17:48
REACTOME_SENSORY_PERCEPTION	0.12040376	605	6.503e-24	7.039e-21	OR5E1:2 OR5IS1:13 OR5A1:125 OR2C1:62 OR52W1:65 OR2S2:66
KIM_ALL_DISORDERS_CALB1_CORR_UP	0.12625427	528	4.157e-23	3.856e-20	HPCA:33 DNMI:64 GRIA2:70 ZFAND5:81 NONO:90 INA:91
REACTOME_NERVOUS_SYSTEM_DEVELOPMENT	0.12081187	570	8.274e-23	6.717e-20	RPS8:11 SPTBN2:36 RPS3:46 RPS17:48 PIK3R1:55 DNMI:64
REACTOME_NEURONAL_SYSTEM	0.14105017	403	3.089e-22	2.229e-19	NLGN3:24 GJD2:41 FLOT1:43 GRIA2:70 GNB2:75 SRC:89
REACTOME_SIGNALING_BY_WNT	0.15355194	322	3.125e-21	2.029e-18	WNT3:4 WNT1:42 GNB2:75 AGO1:88 PFN1:130 WNT10A:147
REACTOME_INFECTIONOUS_DISEASE	0.09069032	933	3.415e-21	5.558e-18	TAF12:5 RPS8:11 ATP1A2:40 MYBPC1:50 TNNC2:52 TNNT3:178 RYR3:212
HSIAQ_HOUSEKEEPING_GENES	0.13763487	381	3.366e-20	1.822e-17	RPS8:11 RPS3:46 RPS17:48 PAX6:56 AAMP:69 PCBP2:85
REACTOME_CELLULAR_RESPONSES_TO_STIMULI	0.09017722	788	1.003e-17	5.009e-15	HI-3-8 RPS8:11 HSPA2:39 RPS3:46 RPS17:48 GSR:57
REACTOME_REGULATION_OF_EXPRESSION_OF_SL	0.18751051	169	4.314e-17	2.001e-14	RPS8:11 RPS3:46 RPS17:48 MAGOH:146 RPL27:167 RPL3L:188
REACTOME_SIGNALING_BY_ROBO_RECEPTORS	0.16603511	214	6.102e-17	2.502e-14	RPS8:11 RPS3:46 RPS17:48 SRC:89 PFN1:130 MAGOH:146
REACTOME_OLFATORY_SIGNALING_PATHWAY	0.12358974	390	6.165e-17	2.502e-14	OR5E1:2 OR5IS1:13 OR5A1:125 OR2C1:62 OR52W1:65 OR2S2:66
REACTOME_SIGNALING_BY_NUCLEAR_RECEPTORS	0.14274502	166	1.084e-16	4.142e-14	PIK3R1:55 GNB2:75 SMC1A:82 AGO1:88 SRC:89 CXCC5:137
REACTOME_EUKARYOTIC_TRANSLATION_INITIAT	0.22046715	118	1.356e-16	4.987e-14	RPS8:11 RPS3:46 RPS17:48 EIF2S1:100 RPL27:167 RPL3L:188
REACTOME_CELLULAR_RESPONSE_TO_STARVATION	0.19330161	153	1.624e-16	5.551e-14	RPS8:11 RPS3:46 RPS17:48 MIOS:68 EIF2S1:100 RPL27:167
REACTOME_MUSCLE_CONTRACTION	0.16495471	201	7.919e-16	2.571e-13	RYR1:10 ATP1A2:40 MYBPC1:50 TNNC2:52 TNNT3:178 RYR3:212
REACTOME_ESR_MEDIATED_SIGNALING	0.15716503	214	2.444e-15	7.558e-13	PIK3R1:55 GNB2:75 SMC1A:82 AGO1:88 SRC:89 CXCC5:137
REACTOME_TRANSMISSION_ACROSS_CHEMICAL_SY	0.13973667	262	7.399e-15	2.184e-12	GRIA2:70 GNB2:75 SRC:89 GABRB3:92 CACNB3:118 ARHGEF9:125
REACTOME_RESPONSE_OF_EIF2AK4_GCN2_TO_AMI	0.22416119	99	1.312e-14	3.705e-12	RPS8:11 RPS3:46 RPS17:48 EIF2S1:100 RPL27:167 RPL3L:188
KEGG_OLFACTORY_TRANSDUCTION	0.11476342	380	1.763e-14	4.770e-12	OR5E1:2 OR5IS1:13 OR5A1:125 OR2C1:62 OR52W1:65 OR2S2:66
DIAZ_CHRONIC_MYELOGENOUS_LEUKEMIA_UP	0.06169833	1356	3.842e-14	9.979e-12	NIPSNAP1:32 SPTBN2:36 CSNK1G3:44 GSR:57 TOP1:67 GNB2:75
GOBERT_OLIGODENDROCYTE_DIFFERENTIATION_U	-0.09352698	561	4.273e-14	1.067e-11	MIS18BP1:31 UNLS4:45 LARP7:88 CENPF:103 CDCA2:111 HAUS6:115
REACTOME_METABOLISM_OF_RNA	0.08395574	691	6.586e-14	1.584e-11	RPS8:11 RPS3:46 RPS17:48 PCBP2:85 CPSF6:87 SNRPD:198
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACT	-0.13550912	258	7.142e-14	1.656e-11	AMHR2:144 TNFSF10:149 TNFRSF11A:204 IL6R:262 IFNAR2:320 TNFRSF19:348
KEGG_RIBOSOME	0.23339605	85	1.037e-13	2.321e-11	RPS8:11 RPS3:46 RPS17:48 RPL27:167 RPL3L:188 RPL32:231
REACTOME_EUKARYOTIC_TRANSLATION_ELONGATI	0.22109341	92	2.353e-13	5.093e-11	RPS8:11 RPS3:46 RPS17:48 RPL27:167 RPL3L:188 RPL32:231
REACTOME_NONSENSE_MEDIATED_DECAY_NMD	0.19901012	113	2.706e-13	5.669e-11	RPS8:11 RPS3:46 RPS17:48 MAGOH:146 RPL27:167 RPL3L:188
REACTOME_SRP_DEPENDENT_COTRANSLATIONAL_P	0.20045121	111	3.034e-13	6.158e-11	RPS8:11 RPS3:46 RPS17:48 RPL27:167 RPL3L:188 SPCS1:205
WP_CYTOSOLASMIC_RIBOSOMAL_PROTEINS	0.22571739	86	4.708e-13	9.264e-11	RPS8:11 RPS3:46 RPS17:48 RPL27:167 RPL3L:188 RPL32:231
FLORIQ_NEOCORTEX_BASAL_RADIAL_GLIA_DN	-0.15346232	186	5.505e-13	1.051e-10	MIS18BP1:31 BRCA1:33 CENPF:103 CDCA2:111 XRC2:117 CDCA3:140
ENK_UV_RESPONSE_KERATINOCYTE_UP	0.09245441	519	6.500e-13	1.206e-10	HI-3-8 ACTN4:26 HPCA:33 RPS3:46 ARHGDIA:61 GNB2:75
REACTOME_TCF_DEPENDENT_SIGNALING_ILN_RESP	0.13832971	226	8.191e-13	1.478e-10	WNT3:4 WNT1:42 H2AX:247 PSDM1:318 HI-3B:370 SFRP1:406
REACTOME_BETA_CATENIN_INDEPENDENT_WNT_SI	0.16987030	145	1.703e-12	2.989e-10	WNT1:42 GNB2:75 AGO1:88 PFN1:130 PRKCG:217 PSDM1:318
REACTOME_TRANSLATION	0.12051563	290	1.771e-12	3.027e-10	RPS8:11 RPS3:46 RPS17:48 EIF2S1:100 RPL27:167 MRPL35:174
REACTOME_RUNX1_REGULATES_TRANSCRIPTION_O	0.18145692	125	2.505e-12	4.171e-10	H2AX:247 PSDM1:318 HI-3B:370 LMO2:425 PSMB4:435 H2BC4:449
MEBARK1_HCC_PROGENITOR_FZD8CRD_UP	-0.08920626	526	2.906e-12	4.718e-10	MIS18BP1:31 BRCA1:33 FANCB:35 NLRP5:61 ARHGAP31:95 BARD1:107

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Congenital myopathy (disorder)	0.13254197	109	1.797e-06	3.853e-03	RYR1:10 MYH7B:37 MYBPC1:50 SLC25A4:141 TNNT3:178 HACD1:207
Graft-vs-Host Disease	-0.09518022	212	1.891e-06	3.853e-03	DNMT1:143 TNFSF10:149 VDR:191 FANCD2:321 SPP1:364 CTSH:436
Nonsyndromic Deafness	0.15711543	79	1.407e-06	3.853e-03	POU4F3:16 POU3F4:21 MYO7A:236 SIX1:324 CDH23:330 GPM2:422
Rheumatoid Arthritis	-0.03621231	1626	1.922e-06	3.853e-03	IRAK1:2 ADAMT57:13 PDE3A:19 ADM:38 PTGER4:96 IRL1:126 SHBG:53
Sarcoidosis	-0.09615545	207	1.962e-06	3.853e-03	ZSCAN12:121 TRH:172 VDR:191 SPP1:364 IGFBP4:513 CRI1:548
Psoriasis	-0.05144113	734	2.577e-06	4.219e-03	TCAIM:9 PDE3A:19 BANK1:73 IRL1:126 TNFSF10:149 IRGM:151
Chromosome Breakage	-0.20488588	40	7.382e-06	7.904e-03	BRCA1:33 FANCB:35 XRC2:117 FANCF:160 BRCA2:201 FANCD2:321
Decreased platelet count	-0.10745897	147	7.128e-06	7.904e-03	SALL4:15 FANCB:35 XRC2:117 ACD:125 FANCF:160 BRCA2:201
Generalized hypotonia	0.05362279	603	8.048e-06	7.904e-03	SYNGAP1:12 ZIC1:22 DNMI:64 ARID1A:73 NONO:90 GABRB3:92
Measles	-0.11864311	119	8.025e-06	7.904e-03	IRF3:178 VDR:191 STAT2:280 IFNAR2:320 CSF3:486 SQTAL1:565
Arthritis	-0.05406056	575	1.087e-05	8.612e-03	MEPE:6 ADAMT57:13 CD248:52 BANK1:73 EGLN1:148 TNFSF10:149
Epilepsy	0.03968480	1114	1.055e-05	8.612e-03	SYNGAP1:12 KIF5C:18 ATP1A2:40 SV2A:54 PAX6:56 DNMI:64
Inflammatory Bowel Diseases	-0.04565009	817	1.140e-05	8.612e-03	BRCA1:33 BANK1:73 TBC1D9:79 IL16:87 IAPP:91 PTGER4:96
Asthma	-0.03786449	1184	1.546e-05	9.489e-03	IRAK1:2 ADM:38 GAB1:63 GFRA4:66 MZ1:65 IL16:87
Behcet Syndrome	-0.08231303	236	1.400e-05	9.489e-03	IRF1:126 VDR:191 MCB:390 IL12B:431 TNFSF13B:595 IRF8:784
Schizophrenia	0.03188366	1748	1.520e-05	9.489e-03	SYNGAP1:12 MPP2:114 SPTBN2:36 GJD2:41 WNT1:42 FLOT1:43
Celiac Disease	-0.07496565	278	1.808e-05	9.865e-03	TBC1D9:79 ARHGAP31:95 IRF1:126 VDR:191 INS-IGF2:231 PML:257
Epileptic encephalopathy	0.05928802	450	1.808e-05	9.865e-03	SYNGAP1:12 KIF5C:18 ATP1A2:40 DNMI:64 SMC1A:82 GABRB3:92
Aplastic Anemia	-0.10366560	141	2.210e-05	1.089e-02	PIGA:120 ACD:125 TNFSF10:149 TET2:12 IL27RA:232 PRSS27:286
Autoimmune Disease	-0.04086358	945	2.611e-05	1.089e-02	ADM:38 NLRP5:61 BANK1:73 TBC1D9:79 IL16:87 PIGA:120
hypopigmented skin patch	-0.14705744	69	2.430e-05	1.089e-02	FANCB:35 XRC2:117 FANCF:160 BRCA2:201 PPOX:261 FANCD2:321
Leprosy	-0.10632212	133	2.362e-05	1.089e-02	IRGM:151 VDR:191 PRKN:258 CCDC122:307 MCB:390 IL12B:431
Primary microcephaly	-0.11607814	110	2.658e-05	1.089e-02	FANCB:35 LARP7:88 CENPF:103 FANCF:160 CEP152:163 BRCA2:201
Respiratory Syncytial Virus Infections	-0.11808454	107	2.495e-05	1.089e-02	IRF1:126 IRF3:178 VDR:191 SPP1:364 MCB:390 DCPIB:1